

RETAIL INVENTORY ANALYTICS USING GOOGLE COLAB

INTRODUCTION

Inventory management is an important aspect of retail business operations. Effective inventory control helps organizations maintain optimal stock levels, reduce shortages, avoid overstocking, and improve customer satisfaction. Data analytics techniques can be used to analyze inventory performance and support business decision-making.

This project focuses on analyzing a retail inventory dataset using Python libraries in Google Colab. Various data analysis and visualization techniques are applied to identify sales trends, low-stock products, supplier contributions, and inventory performance.

OBJECTIVES

- Analyze category-wise sales performance.
- Identify products approaching reorder levels.
- Evaluate supplier contribution to inventory.
- Study the relationship between stock availability and sales.
- Generate business insights and recommendations for inventory management.

TOOLS AND TECHNOLOGIES USED

- Google Colab
- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn

DATASET INFORMATION

The retail inventory dataset contains information about products, categories, suppliers, stock levels, sales, and reorder points.

Dataset Summary:

- Total Records: 1000
- Total Columns: 11
- Missing Values: 0

- Duplicate Records: 0

Important Attributes:

- Product ID
- Product Name
- Category
- Supplier Name
- Unit Cost
- Retail Price
- Opening Stock
- Units Sold
- Closing Stock
- Reorder Point
- Stock Status

DATA CLEANING

The following data cleaning activities were performed:

- Checked for missing values.
- Checked for duplicate records.
- Validated stock-related fields.
- Verified dataset consistency.

The dataset was found to be clean and suitable for analysis.

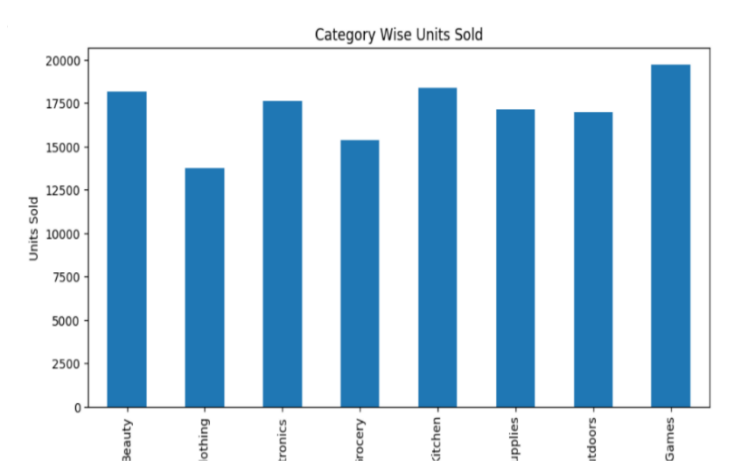
EXPLORATORY DATA ANALYSIS

The following analyses were performed:

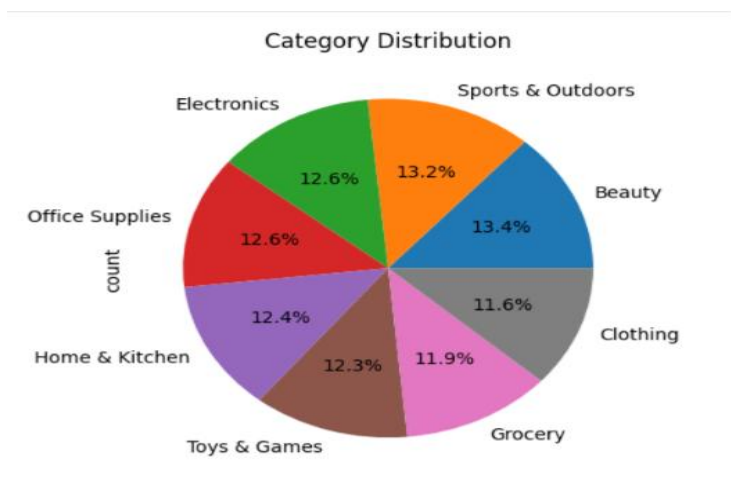
- Category-wise Product Count
- Category-wise Units Sold
- Top 10 Best-Selling Products
- Top 10 Low-Stock Products
- Supplier-wise Inventory Distribution

Visualizations Created:

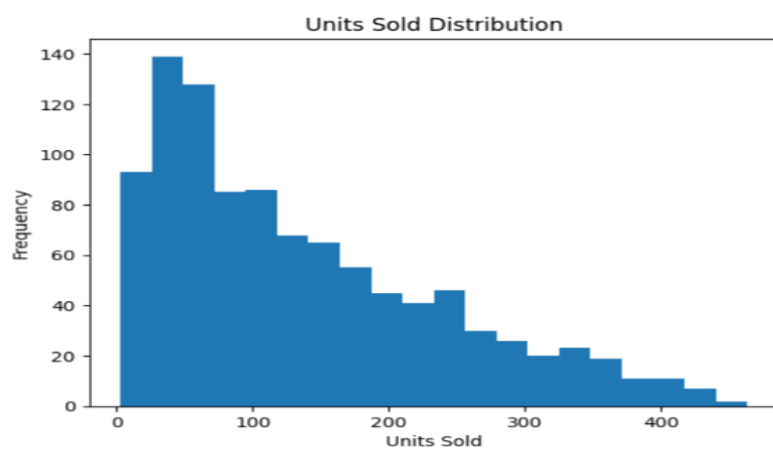
- Bar Chart



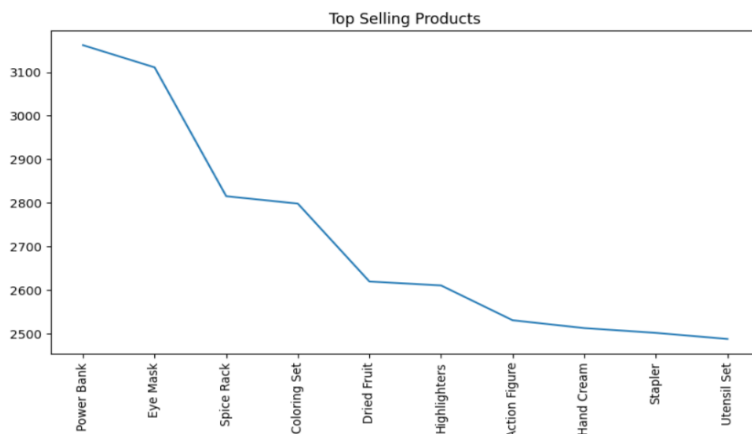
• Pie Chart •



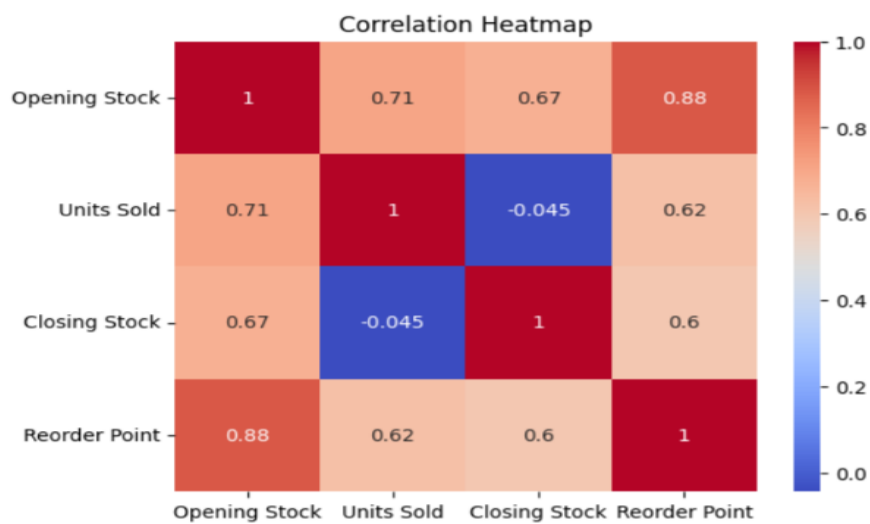
• Histogram



- Line Graph



- Correlation Heatmap



BUSINESS INSIGHTS

- Low-stock products require immediate replenishment.
- Toys & Games is one of the highest-selling categories.
- Electronics products show strong sales potential.
- Top-selling products should maintain sufficient stock.
- Supplier dependency risk exists due to reliance on a few suppliers.

RECOMMENDATIONS

- Replenish low-stock products.
- Maintain safety stock for fast-moving items.

- Reduce dependency on a few suppliers.
- Monitor inventory regularly.
- Focus on high-performing categories.

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CONCLUSION

The Retail Inventory Analytics project successfully analyzed inventory data using Python and Google Colab. The study identified high-performing categories, best-selling products, low-stock items, and supplier contributions. Data visualizations provided meaningful insights into inventory trends and sales performance.

The findings can help management improve inventory planning, reduce stock shortages, optimize supplier management, and enhance overall business efficiency. The project demonstrates how data analytics can support effective inventory management and informed business decision-making.